

Project ATHENA — Day 5

Theme: Returns, Volatility & Compounding — The Mathematics of Wealth

"Price tells you what happened. Returns tell you how much you actually earned."

Objective: Learn how professional investors measure performance and why volatility is the "price you pay" for returns.

Time Required: ~40 minutes

Today's Big Idea

Imagine these two investments.

	Stock Start	End
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A	₹100	₹110
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B	₹1,000	₹1,010
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Which performed better?

Most beginners instinctively say **Stock B**, because it gained ₹10 instead of ₹10? Wait—both gained ₹10 in absolute terms.

But returns tell a different story:

- **Stock A:** +10%
- **Stock B:** +1%

This is why quants almost never compare price changes directly. They compare **returns**.

Step 1 — Understanding Returns

The simplest measure is the **simple return**:

$$\text{Return} = \frac{\text{Current Price} - \text{Previous Price}}{\text{Previous Price}}$$

Example:

Yesterday: ₹500

Today: ₹525

Return = $(525-500)/500$

= 5%

Everything in ATHENA—performance, strategy evaluation, portfolio analysis—will ultimately be based on returns, not prices.

Step 2 — The Magic (and Danger) of Compounding

Suppose you earn:

10%

every year.

Starting capital:

₹100,000

After one year:

₹110,000

After two years:

Not ₹120,000.

Instead:

₹121,000

because you're earning returns on your previous gains.

Now imagine:

15%

for 20 years.

The difference between simple growth and compounded growth becomes enormous.

One of the biggest mistakes new traders make is chasing very high returns while ignoring the power of steady compounding.

Step 3 — Volatility

Imagine two portfolios.

Portfolio A

+1%

+2%

+1%

-1%

+2%

Portfolio B

+15%

-12%

+20%

-18%

+10%

Both may produce similar average returns.

Which would you rather own?

Most long-term investors prefer **Portfolio A**, because the journey is smoother.

Volatility measures **how much returns fluctuate**.

High returns with extreme volatility can be psychologically and financially difficult to sustain.

Step 4 — Why Losses Hurt More Than Gains Help

Suppose your portfolio loses:

50%

To recover, you need:

100%

Not 50%.

Here's a useful mental table:

Loss Gain Needed to Break Even

10% 11.1%

20% 25%

30% 42.9%

40% 66.7%

50% 100%

This is why ATHENA will prioritize **drawdown control** alongside return generation.

Engineering Insight

In machine learning, we don't judge a model by a single prediction.

We evaluate it over many observations using metrics.

Trading is the same.

Never judge a strategy based on one trade.

Judge it over hundreds—or ideally thousands—of trades.

The goal is to maximize **expected performance over time**, not to win every individual trade.

Practical Exercise (≈40 Minutes)

Part 1 — Build ATHENA's Return Calculator (25 min)

Create:

`src/performance/returns.py`

Implement functions for:

`simple_return(previous_price, current_price)`

```
cumulative_return(price_series)
```

```
daily_returns(price_series)
```

Keep the implementation clean and well-documented.

Part 2 — Test It (10 min)

Write unit tests covering:

- Positive returns
- Negative returns
- Zero return
- Multiple-day cumulative return

Think about edge cases such as division by zero or missing values, even if you don't implement full handling yet.

Part 3 — Journal (5 min)

Answer:

If two strategies produce the same annual return, but one experiences half the volatility of the other, which would I trust more and why?

There isn't a single correct answer—but your reasoning matters.

Architecture Update

ATHENA now grows to:

```
src/
```

```
|— ingestion/
```

```
|— cleaning/
```

```
|— features/
```

```
|— indicators/
```

```
|— performance/
```

```
|   └─ returns.py
|   └─ strategies/
|   └─ risk/
|   └─ portfolio/
└─ backtesting/
```

Notice how we're building reusable modules instead of isolated scripts.

Review Question

Two strategies each earn **20%** over a year.

Strategy A:

- Maximum drawdown: 8%
- Moderate volatility

Strategy B:

- Maximum drawdown: 35%
- Very high volatility

Which would generally be more attractive for long-term deployment?

Answer: Strategy A.

In quantitative investing, achieving the same return with less risk is usually preferable.

Looking Ahead — Day 6

Tomorrow we'll begin working with one of the most important concepts in technical analysis and quantitative research:

Moving Averages: Finding the Market's Trend

You'll learn:

- Simple vs. Exponential Moving Averages
- Why moving averages reduce noise
- Their strengths and limitations
- How to implement them efficiently in Python
- How they'll become the first feature in ATHENA's strategy engine

Mentor's Note

Today's lesson may have seemed mathematical, but it's foundational. Over the coming months, almost every performance metric we'll use—Sharpe Ratio, Sortino Ratio, Maximum Drawdown, CAGR, Calmar Ratio, rolling volatility, Value at Risk—builds on the ideas you learned today.

One small suggestion for the project architecture: as ATHENA grows, consider treating it like a production-grade Python package from the beginning. Use type hints, docstrings, consistent formatting, and automated tests for every new module. Those habits will pay off far more than squeezing out a few extra percentage points in a backtest.